

SAI POOJITHA SAJJAVARAPU

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Education

Rajiv Gandhi University of Knowledge and Technologies, Nuzvid IIIT
Bachelor of Technology in Computer Science, CGPA-8.5

Nuzvid, Eluru
September 2022 - April 2026

Experience

Quantum Gandiva AI

Onsite (November 2025 – Present)

AI Backend Engineer & Data Engineer

Visakhapatnam

- Built the **entire backend stack** for a production Agentic AI product, handling agent orchestration, execution logic, and system coordination.
- Developed and shipped **FastAPI-based backend services** with asynchronous request handling, modular API design, and optimized data flow to integrate LLM reasoning, agent memory, and dynamic tool-calling workflows.
- Implemented **AWS-backed data pipelines** to support real-time ingestion and decision-making with low operational overhead.

Parabola9

Onsite (December 2024 - May 2025)

AI/ML Developer

Nuzvid, Eluru

- Developed and optimized advanced **machine learning models** to solve complex problems, creating impactful AI solutions, including **generative AI applications** and **LLM-based systems**.
- Deployed **AI models** in production with scalable integration, and continuously monitored and refined their performance to ensure efficiency and adaptability.
- Collaborated with engineers and domain experts to design, integrate, and **deploy AI models**, improving system efficiency and ensuring seamless end-to-end application performance

Projects

Tailor-Fit: AI-Powered Shopping Platform 🛒

- Tailor-Fit is an innovative fashion-tech platform that bridges the gap between the precision of traditional tailoring and the convenience of online shopping.
- Designed and integrated an end-to-end ML pipeline for body measurement estimation and virtual try-on, ensuring scalable and real-time inference for user requests.
- Automated measurement extraction and tailoring workflows, achieving a 35% simulated return-rate reduction and improved precision fitting via a connected local tailor network.
- **Tech Stack:** React.js, Node.js, Express.js, Python, TensorFlow, PyTorch, PiHuD, HR-VITON

AI-Driven Blood Donation Network: Intelligent Blood Donor-Bank Matching System 🩸

- Developed an **AI-driven blood donation platform** integrating **FastAPI (backend)** and **Flutter (frontend)** for real-time donor-blood bank connectivity.
- Implemented **intelligent donor-bank matching** using **Vertex AI**, **LangChain**, and **FAISS vector search** to enhance accuracy and reduce response latency.
- Integrated **Google Maps API** with geofencing and smart alerts for **location-aware donor mobilization** and emergency blood request handling.
- **Tech Stack:** FastAPI, Flutter, Firebase, Vertex AI, LangChain, FAISS, Google Cloud.

MediBuddy: AI-Powered Prescription Interpreter 🩺

- Developed a web application that interprets medical prescriptions and simplifies terminology using **Gemini LLM API via backend API integration**. Focused on accessibility and usability for elderly users by simplifying medical language.
- Designed a modular, responsive UI with **React.js** and managed state using **React Hooks** and **Context API**.
- Implemented **secure data handling**, **voice output for summaries**, and **error-resistant API integration** for enhanced user experience.
- **Tech Stack:** React.js, Tailwind CSS, Vite, **Gemini API**, PostCSS, ESLint, JavaScript, **Docker**

Technical Skills

- **Programming Languages:** Java, C, C++, Python, JavaScript, Dart
- **Technologies & Frameworks:** Node.js, **FastAPI**, Django, REST APIs, **Agentic AI**, LLMs, **Generative AI**, NLP, **Machine Learning**, **Deep Learning**, Flutter, ReactJS
- **Tools & DevOps:** Git, Docker, Postman, GitHub Actions, CI/CD, VS Code, **Redis (Streams)**, Kubernetes, Terraform
- **Databases & Cloud Platforms:** MySQL, PostgreSQL, AWS (EC2, S3, Lambda, RDS, VPC), GCP, Azure

Achievements

- **Smart India Hackathon 2024 Winner:** Mentored a winning team that secured **1st Prize** at a prestigious national hackathon by developing an AI-based solution to address **bus bunching** in public transportation.
- **Teczite Mega Expo 1st Prize Winner:** Secured **1st Prize** at a national-level expo by developing an **AI-powered security surveillance system** integrating **gesture recognition**, **YOLO-based anomaly detection**, and **real-time video captioning**.
- **IIIT Hackathon Winner – Parabola9:** Won **1st Prize** for developing **VideoGPT**, a deep learning-based tool for generating video captions using frame optimization.
- **Problem-Solving Expertise:** Solved over **500+ LeetCode problems** with a max contest rating of **1600+**.